



Parking occupancy prediction under COVID-19 anti-pandemic policies: A model based on a policy-aware temporal convolutional network

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ABSTRACT

Real-time and reliable parking occupancy prediction is critical for managing transportation infrastructure and services. However, the COVID-19 pandemic and its corresponding response have led to changes in daily routines and behaviors, including travel time distribution, frequency, and parking time. Changes in parking behavior pose significant challenges for accurately predicting parking occupancy. To address this issue, we proposed a novel policy-aware temporal convolutional network (P-TCN) model that considers the policy dependence of time-series data irregularities. The model employed 3D vector data as input to predict future parking occupancy while ensuring reliability through 10-fold cross-validation. The SHapley Additive exPlanations (SHAP) method was also utilized to analyze the impact of the feature variables on the model. Experimental results show that the P-TCN model accurately identifies the effects of anti-pandemic policies and outperforms existing models such as multiple linear regression (MLR), autoregressive integrated moving average (ARIMA), fully convolutional network (FCN), long short-term memory (LSTM), and temporal convolutional network (TCN). Moreover, the feature variables “policy”, “weather”, and “time information” contribute significantly to the model prediction results, with “policy” having the most significant effect. This model has wide applicability for predicting and interpreting parking occupancy in different countries and regions. As the COVID-19 pandemic gradually normalizes, the P-TCN model’s parking occupancy information will help managers better understand residents’ behavior and respond flexibly to emergencies.

1. Introduction

Challenges in parking have long plagued drivers and traffic managers, significantly affecting the development of urban

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transportation (Shao et al., 2020; Yan et al., 2020). In urban cores, drivers often spend considerable time cruising for parking spaces because of the deepening conflict between supply and demand (Shao et al., 2016; van Ommeren et al., 2011). Although large parking facilities are often available in residential areas in many densely populated cities, these parking spaces are typically reserved for residents (Liang et al., 2019). However, the often overlooked fact is that parking resources in residential areas can also help solve parking problems (Jiang and Fan, 2020; Zhang et al., 2020b). As residents generally have morning and evening commuting patterns, parking spaces are often vacant during working hours (Lai et al., 2021; Su and Wang, 2020). Therefore, utilizing these vacant residential parking spaces can provide convenient parking for drivers heading to facilities near residential areas and can serve as an effective supplement to public parking (Niu et al., 2021). Furthermore, this approach plays a crucial role in effectively mitigating the adverse effects of insufficient parking spaces and unreasonable cruising on urban traffic (Ning et al., 2021; Shao et al., 2020).

The absence of a reliable source of information on current and predicted parking occupancy poses a challenge in improving the capacity of residential parking facilities to provide parking services to visiting drivers (Wang et al., 2022; Yang and Qian, 2019). While parking management and guidance system developments and applications have attempted to address this issue, accurate and reliable prediction of future parking occupancy in residential areas is not straightforward (Gong et al., 2021). Factors such as weather and time information affect parking behavior, leading to an irregular temporal distribution of parking behavior (Su and Wang, 2020; Xiao et al., 2020). For example, slow traffic due to inclement weather may cause residents to change their travel times and modes, resulting in recent temporal changes in parking occupancy (Wu and Liao, 2020). During these times, residents typically have different travel plans and modes, leading to a different distribution of parking times than usual for residents parking in residential areas (Bursa et al., 2022; Han et al., 2020). Therefore, many studies consider weather and time information as key factors in understanding complex parking-related behaviors and often employ artificial neural networks (ANNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs) for parking occupancy prediction (Islam et al., 2022; Lai et al., 2021).

However, relying solely on weather and time information to establish a model makes it susceptible to unexpected events that result in significant changes in time patterns, leading to less accurate predictions (Pereira et al., 2022). The COVID-19 pandemic has seen the implementation of various non-pharmaceutical interventions (NPIs) and measures worldwide to curb the virus's spread in recent years (Chen et al., 2014; De Palma et al., 2022). Pandemic responses, such as stay-at-home orders, lockdowns, social distancing practices, and travel restrictions, severely restrict citizens' travel and work behavior. Many companies and institutions have adopted flexible, remote working arrangements. However, overly flexible working hours can lead to scattered and uncertain resident travel times, making it difficult to track irregular travel behavior (Muren et al., 2022). This presents a significant challenge for guaranteeing the accuracy of predicting residential parking occupancy (Ye et al., 2023). Previous research focused on refining the model structure to improve parking predictions, but it has largely disregarded the significant challenges posed by the inability to adequately capture irregular parking behavior for accurate parking occupancy prediction (Xiao et al., 2020; Xu et al., 2016; Yang and Qian, 2017). Therefore, a more in-depth study of parking behavior under irregular conditions, particularly the influence of the COVID-19 pandemic response, is necessary to enhance the predictive effectiveness of a model.

The objective of this research is to overcome the limitations of existing studies in accurately and stably predicting time-series irregular parking occupancy by developing a P-TCN model. The P-TCN model integrates the time and policy features captured from multiple data sources to predict future parking occupancy under irregular states. This study contributes to the literature on parking forecasting in three ways. First, it extensively considers the time-series variability induced by the COVID-19 pandemic response on parking behavior to accurately predict parking occupancy in irregular states. Moreover, historical information on vehicles entering and leaving residential parking facilities is used to update and predict future parking occupancy situations. Second, this study addresses the challenge of predicting nonregular parking behavior in residential areas using the P-TCN model. The variability of time-series policies for parking occupancy is analyzed and characterized, improving the model's predictive performance and bridging the gap in identifying policy variability in existing research. Finally, a TCN model with long-lasting memory preservation characteristics and causal feature effects is applied to predict parking occupancy. This ensures that the model identifies data features on a longer timescale without causing information leakage induced from the future to the past.

The remainder of this paper is organized as follows. Section 2 provides a literature review of related studies on parking occupancy prediction. Section 3 proposes an occupancy prediction model that considers the time-series irregularity of residents' parking behavior and integrates policy features. Section 4 presents the results of parking occupancy prediction based on historical data compared with the prediction effects of different methods, followed by a discussion of the major results. Finally, the key findings are summarized in Section 5, and proposals are made for future research.

2. Literature review

2.1. Parking demand

Parking demand has received considerable attention in the literature on individual travel behavior (Nian et al., 2020; Yan et al., 2019). Previous research has focused on the impact of changes in parking prices on travel behavior and parking demand, particularly on issues related to free and low-priced curb parking (Millard-Ball et al., 2014; Yan et al., 2019). Kelly and Clinch (2009) estimated the elasticity of demand for curb parking prices based on revealed preference data, while Cats et al. (2016) empirically measured the effects of parking pricing policies on curb parking behavior based on parking transaction data. The results indicate that drivers are sensitive to parking prices, and there is a significant negative correlation between parking demand and parking prices (Mo et al., 2021; Ottosson et al., 2013). Similar conclusions have been drawn in several modeling-based studies (Ornelas et al., 2023; Shao et al., 2016). Cheaper and more convenient off-street parking usually attracts more demand for parking, resulting in a high turnover and occupancy

of parking spaces (Chen et al., 2016; Gragera and Albalade, 2016). Once the target value of curb parking occupancy exceeds 85%, as recommended by Shoup (2006), the increase in cruise time versus parking occupancy is superlinear (Abbott and Bigazzi, 2017; Geva et al., 2022). Therefore, understanding drivers' responses to parking availability and prices is the key to establishing sustainable parking policies (Fulman et al., 2020; Xiao et al., 2018a). However, as long as curb parking prices remain lower than off-street parking prices, cruising for parking will continue (Millard-Ball et al., 2014). To reduce cruising for parking, many cities have adopted a strategy based on Shoup (2006)'s research, which involves raising curb parking prices to match off-street parking prices (Cats et al., 2016; Kelly and Clinch, 2009; Ottosson et al., 2013).

Although raising curb parking prices can reduce cruising behavior, the shortage of parking spaces in densely populated cities still makes it difficult for drivers to find suitable spaces, even if curb and off-street parking prices are consistent (Abbott and Bigazzi, 2017). Insufficient public parking spaces have become a major cause of parking difficulties (Huang et al., 2021; van Ommeren et al., 2011). However, building multiple parking facilities in a short period is unrealistic, as new parking demands will offset this impact in the near future (Niu et al., 2021). To alleviate parking difficulties, allowing travelers to use vacant private parking spaces during working hours has attracted considerable attention from researchers and government agencies (Xu et al., 2016; Yan et al., 2020).

2.2. Parking occupancy prediction

The real-time and reliable prediction of parking occupancy is an important input module for studying traffic management, such as shared parking and shared mobility systems (Chen et al., 2014; Rodier and Shaheen, 2010). However, many studies on parking occupancy rates are based on objective estimates of the regularity of the current moment sample to the overall sample, rather than the prediction of near-term parking occupancy rates (Rodier and Shaheen, 2010; Yang and Qian, 2019). Estimation methods such as Markov and Monte Carlo models have shown limited effectiveness in explaining the nonlinearity and heterogeneity of many influencing factors (Pak et al., 2020; Park et al., 2016; Xiao et al., 2018b). Meanwhile, parking behavior, pricing, and policy studies require a fine-grained prediction of changes in parking occupancy over time (Yang and Qian, 2017). Private car parking is also influenced by exogenous variables such as weather and time information (Nguyen-Phuoc et al., 2018; Pierce and Shoup, 2013). Traditional regression techniques may handle multiple data sources, but their ability to capture relationships between data is limited by the volume and speed of data processing (Kim et al., 2020; Ma et al., 2017). Moreover, traditional statistical regression models have limitations when dealing with the veracity issues of large datasets. Traditional regression models are sensitive to data anomalies and outliers and struggle to control the effects of various factors that affect their accuracy (Anscombe, 1973; Cook and Weisberg, 1982; Hastie et al., 2009; Pendyala, 2018).

To address these challenges, statistical learning methods driven by large amounts of historical data have been developed, including support vector machines, random forests, ANNs, and other nonlinear models (Ahangari et al., 2021; Mokhtarimousavi et al., 2019; Pirra and Diana, 2019; Wang and Ross, 2018). These models handle complex data veracity and causality problems and improve the predictive performance by expanding the assumption space (Gao et al., 2022). In particular, nonlinear models outperform linear models in achieving high-accuracy predictive performance.

In recent years, deep learning methods, such as RNNs, have significantly outperformed traditional statistical and machine learning methods, becoming crucial in time series modeling (Kumar et al., 2022; LeCun et al., 2015). RNNs have found applications in the transportation field, particularly in traffic flow prediction, traffic congestion identification, and parking occupancy prediction, owing to their ability to learn the temporal dependencies between traffic features from historical data (Ma et al., 2015; Yang et al., 2022). However, RNN models may exhibit limited prediction performance when dealing with irregular or complex time series data (Wen et al., 2019).

TCNs derived from CNN architectures have shown excellent performance in sequence prediction tasks (Del Pizzo et al., 2021). TCNs integrate RNN sequence modeling capabilities and the parallel processing feature capabilities of CNNs (Zhang et al., 2022b). In parking occupancy prediction tasks, TCNs can learn temporal dependencies between different features and apply these patterns to predict future parking occupancy situations (Liu et al., 2021). Furthermore, the strong parallel processing capability of TCN models allows them to handle multiple features simultaneously, thereby enhancing their efficiency and prediction performance (Liang et al., 2023a; Zheng et al., 2021). TCN structures have been shown to outperform RNNs and their variants in terms of prediction accuracy and memory retention in various sequence-modeling tasks (Yin et al., 2017). In addition, the simpler structure of TCNs makes them easier to train (Guo and Yuan, 2020; Ren et al., 2022). The causal features inherent in causal convolution, a core structure in TCNs, may further aid in parking occupancy prediction (Yuan et al., 2014). As a result, TCNs have significant advantages in handling time-series data for residential parking occupancy (Liang et al., 2022b).

TCNs have been applied to air pollution prediction (Zhang et al., 2021c), traffic flow prediction (Zhang et al., 2021d), and travel demand prediction (Zhang et al., 2021b; Zhang et al., 2020a). Despite these successful applications in transportation, the use of TCN models for parking occupancy prediction in residential areas remains underexplored. Therefore, it is essential to investigate whether TCNs can serve as effective alternative models for parking occupancy prediction.

2.3. COVID-19 pandemic effects on parking behavior

The COVID-19 pandemic and related pandemic response measures have significantly impacted residents' parking behavior, as shown in related studies (Fokker et al., 2022; Ye et al., 2023). Changes in travel behavior, including travel time distribution, frequency, and parking time, have become more flexible and random, making it difficult to accurately predict using traditional time-series machine learning methods (Ning et al., 2021). Moreover, many residents have shifted to safer and more private modes of travel,

such as walking or cycling, making it challenging to adapt previous travel behavior prediction methods based on transportation flow analysis to new travel modes and preferences (Parker et al., 2021). Although several studies have attempted to construct new parking prediction models during the COVID-19 pandemic, these models typically do not consider the impact of pandemic response measures on parking behavior, making the accuracy and reliability of these models challenging (Liu et al., 2021; Wang et al., 2022). Gong et al. (2021) used a machine learning approach to compare the prediction performance of parking occupancy before and during the COVID-19 outbreak and found a significant decrease in the prediction performance. Therefore, given the uncertainty and changes in parking behavior during the COVID-19 pandemic, it is essential to adopt parking occupancy prediction methods that can identify policy implications.

In summary, the P-TCN model proposed in this study offers a novel approach with policy-aware attributes that can efficiently capture policies and time-series features from data and accurately predict parking occupancy in residential areas with high temporal resolution. The application of this model can provide policymakers and managers with more accurate parking demand information, enabling them to formulate more precise pandemic response measures, promote scientific and refined traffic management, and improve the safety and efficiency of urban traffic management.

3. Methodology

In this section, we introduce the prediction process into three parts: (1) policy variability identification, (2) model structure design, and (3) model prediction and evaluation. First, we analyze and categorize the differences in residents' parking behavior based on the COVID-19 pandemic response issued by the State Council of the People's Republic of China. Second, we propose a P-TCN structure that builds on and extends the existing TCN prediction structure and is tested for parking occupancy prediction. Finally, we use the rectified linear unit (ReLU) function as the activation function of the prediction model to complete the prediction by connecting all features and feeding them into the regression layer. The model is evaluated based on the prediction results. In addition, a 10-fold cross-validation approach is used to ensure the reliability of the data samples and to obtain a better generalization ability. The following subsections provide further description of the methodology in each stage. Fig. 1 illustrates the structural framework of the study.

3.1. Dataset

This study collected 52,128 parking occupancy-related datasets from November 1, 2021, to April 30, 2022, over 181 days. The datasets included baseline and weather features from three residential parking garages located in Harbin, the capital city of Heilongjiang Province, China. Each parking garage had 17,376 data points collected; Fig. 2 shows their locations in the city core. Residential areas 1, 2, and 3 had parking garages with 940, 860, and 1,045 parking spaces, respectively.

The target variable for this study was parking occupancy data, whereas the feature variables were derived from statistical observations based on parking data. These included baseline features such as time, day of the month, day of the week, holidays, the type of COVID-19 pandemic response, and weather features such as temperature, humidity, snowfall, and precipitation. Parking occupancy data were recorded every 15 min, and weather features were recorded hourly, as described in Table 1.

Notably, residential parking garages are typically located within residential areas and primarily provide parking services to the residents of those areas. Many parking garages use access control systems that can automatically identify residents' license plates, allowing them to enter and exit the garages without needing to stop and present any identification.

As shown in Fig. 3, the average residential parking occupancy profile illustrates that the parking occupancy varies significantly over time. During the day, the parking occupancy rate is relatively low because residents are typically out working and shopping. However,

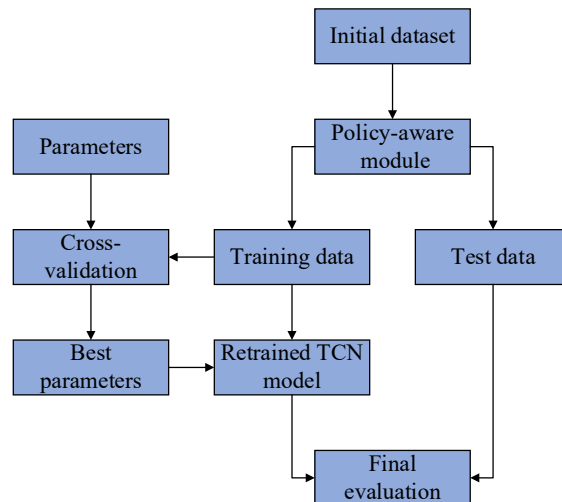


Fig. 1. Training Flowchart of the P-TCN model for parking occupancy prediction.

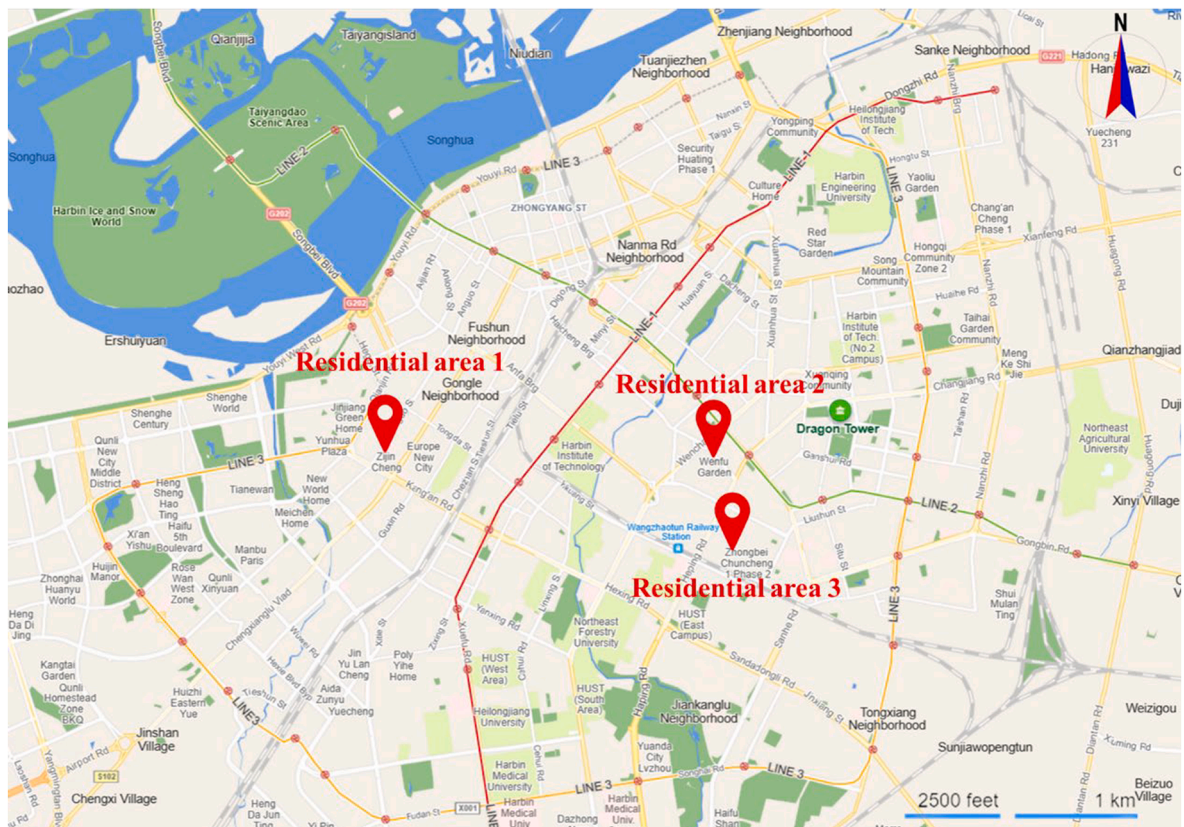


Fig. 2. The location of the three parking garages in the city core.

Table 1

Features related to parking occupancy data.

Category	Variables	Description of features variable
Target variables	Parking occupancy	The actual parking occupancy is recorded every 15 min means that 96 data are collected per parking garage per day. In 6 months, 17,376 data will be recorded per residential parking garage.
Feature variables	Time	Hours and minutes
	Month	Month of the year (0–11)
	Week	Day of the week (0–6)
	Season	0–3 represents four seasons
	Holiday	0 for holidays, 1 else
	Policy	0–2 represent low-, Mid-, and high-risk policies, respectively
	Temperature	Temperature measured every hour (°C)
	Humidity	Humidity measured every hour (%)
	Snowfall	Snowfall measured every hour (mm)
	Precipitation	Precipitation measured every hour (mm)

at night, when most residents return home, the parking occupancy rate increases. Additionally, the parking behavior of residents can vary significantly between weekdays and weekends. Residential parking occupancy is typically lower on weekdays than on weekends because residents use their vehicles more frequently on weekdays to go outside for work, while weekends are typically more leisurely with fewer people going out and thus higher parking occupancy.

3.2. Differentiated identification of policies

It is important to note that the outbreak of COVID-19 and the associated pandemic response had a significant impact on residents' parking behavior, making it more flexible and random, which makes accurate prediction challenging. Therefore, in this study, the authors classified the time-series data according to the pandemic response issued by the State Council of the People's Republic of China, which includes different restrictions on people's travel and activities, physical distancing, and community containment measures. Policies are divided into three levels based on the severity of the pandemic, as shown in Table 2. This approach allows for

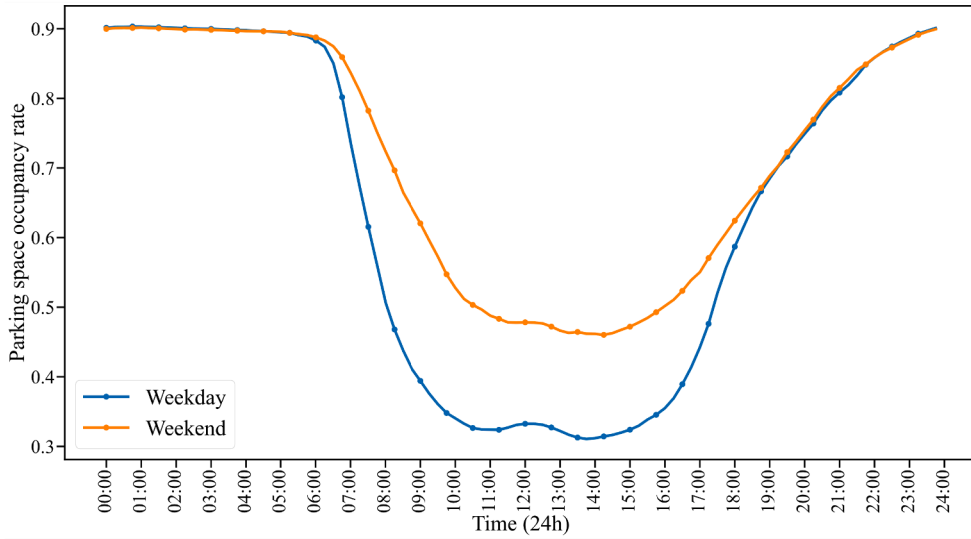


Fig. 3. The average profile of parking occupancy in residential parking garages in 24 h.

better filtering and labeling of characteristic data and considers the irregular state of parking behavior caused by the variability of pandemic-related policies.

The least significant difference (LSD) test developed by Fisher (1935) was used as the policy-aware module of the proposed P-TCN model. The LSD method is a multiple comparison method that is often applied by parking researchers to determine whether there are significant differences between two or more samples (Cobb et al., 2021; Jamson et al., 2008). The method involves first performing variance analysis to determine if there are significant differences and then comparing the differences between the total means of each group to determine if they are significantly different (Larson et al., 2014; Lu et al., 2015). Specifically, the daily average data were divided into seven groups according to the day of the week, from Monday to Sunday. Pairwise comparisons were then performed between each group of data to identify any significant differences and similarities between days of the week. If significant differences in parking behavior were identified, pairwise comparisons were conducted between low-, mid-, and high-risk policies. In addition, to ensure the accuracy of the variance analysis, the sample data were tested for normal distribution (Cobb et al., 2021; O'Hern et al., 2019). The results of the comparative analysis of the target and contrast groups are presented in Section 4.1.

The rationale behind the value of the LSD technique arises from the observation that the value of the t statistic lies in evaluating the differences between groups i and j (Williams and Abdi, 2010). When $i \neq j$, it is equal to

$$LSD = \sqrt{S_l^2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right) \cdot t_{N-k, 1-\frac{\alpha}{2}}} \quad (1)$$

where $t_{N-k, 1-\frac{\alpha}{2}}$ is the critical value of the t distribution obtained by looking up the table of t distributions with degrees of freedom $N-k$; N is the total number of samples; k is the number of different levels in the factor; S_l^2 is the within-group variance; n_i and n_j are the sample sizes of the target and contrast groups, respectively. If the absolute value of the mean difference is greater than that of the LSD, then the decision is made based on the significance level α ($\alpha = 0.05$).

To evaluate the difference between the means of the target group i and contrast group j , the absolute value between the pair of means was compared with the LSD value. The difference between the means is significant when

$$|\bar{Y}_i - \bar{Y}_j| \geq LSD \quad (2)$$

where $\bar{Y}_i$ and $\bar{Y}_j$ are the means of the i and j groups, respectively. This procedure was repeated for all the $\frac{k(k-1)}{2}$ comparisons.

Table 2

The 3 levels of criteria of COVID-19.

Risk Levels	Delineation of Criteria	Anti-Pandemic Policies
Low-risk	Absence of confirmed cases or no new confirmed cases for 14 consecutive days.	Health monitoring and services
Mid-risk	There are new confirmed cases within 14 days, and the cumulative number of confirmed cases does not exceed 50.	Cessation of all commercial activities, but no mandatory requirements for working at home.
High-risk	Cumulative cases exceeding 50 or aggregated outbreaks occurring within 14 days.	Closed community, traffic control, and working at home.

$$P_i = \{|\bar{Y}_i - \bar{Y}_j| \langle LSD \rangle\} \quad (3)$$

Given that not all data from the target and contrast groups were significantly different, data related to parking occupancy with insignificant differences were aggregated and organized into a vector P_i , $i \in 1, 2, 3$, in accordance with the COVID-19 pandemic response. P_i was used as the input feature vector of the model, representing the parking occupancy information of the time series under the i th policy.

Parking occupancy data were classified in accordance with different policies, along with the time step T and feature variable F associated with parking data. When combined, they form a 3D vector $X \in R^{P_i \times T \times F}$ as input features.

3.3. Structure of the proposed network

The proposed P-TCN model is a modification of a TCN that incorporates a policy-aware module, as shown in Fig. 4. The model operates based on two key principles for processing time series problems. First, it ensures that no information leaks from the future to the past, such that the output at time t in the convolution convolves only with the elements at time t or in the previous layer. Second, the model uses a one-dimensional (1D) FCN, which guarantees the same input and output sequence data size for each hidden layer, regardless of the input sequence data size.

Although the TCN has a causal convolution that satisfies backward and forward dependence of the data on linear time series, its coverage of longer historical information is limited because of constraints posed by its capture horizon (Del Pizzo et al., 2021). However, dilated convolution mechanisms allow CNNs to expand their reception field without introducing additional parameters, as shown in Fig. 4(a).

The convolution kernel can be visualized as a small window that scans the input data and extracts features. However, the kernel can only access the adjacent data in the dataset. Dilated convolution, on the other hand, enables the convolution kernel to skip some information by adding intervals, allowing access to more data. By expanding the reception field of the convolution kernel, dilated convolution allows the kernel to capture more data and better model the global information.

Expansion via dilated convolution increases the receptive field of the next layer of neurons in the causal neural network regarding the historical data of the previous neuron. This property improves the ability of causal convolutional networks to model time-series prediction tasks that require longer memory. The proposed causal networks adjust the size of the receptive field by increasing the combined effects of the kernel size k and dilated factor d , where d follows exponential growth with the depth n of the network, i.e., $d = 2^n$, to ensure effective coverage of the long-term history (Bai et al., 2018; Zhang et al., 2021c). Thus, the dilated convolution operation $F(s)$ is applied to a sequence input x using a convolution kernel f of size k . The operation on a sequence element S is defined as

$$F(s) = (x *_d f)(s) = \sum_i^{k-1} f(i) \cdot x_{s-d \cdot i} \quad (4)$$

where $s - d \cdot i$ indicates the past direction and $*_d$ represents the dilated convolution operator, which is distinct from the normal convolution operation. The normal convolution operator ($*$) is a specific version of the dilated convolution when $d = 1$.

Residual networks are essential components of P-TCNs. The simple connections between the layers in the TCN are replaced by the residual structure depicted in Fig. 4(b), which enables the P-TCN to have a deeper and broader network while ensuring network stability and effectively addressing the gradient disappearance problem (Ren et al., 2022; Zhang et al., 2021b). The residual module of the P-TCN consists of a two-layer dilated causal convolution and ReLU (Wen et al., 2019). The output of the dilated causal convolution undergoes weight normalization, which involves adjusting the weights in the network to reduce the complexity of the model and prevent overfitting (Li et al., 2021). The normalized output was then utilized as the input for the ReLU function, denoted as x_{norm} . The ReLU function was used as an activation function for each convolutional layer to improve the nonlinear representation of the model by setting the threshold to 0 (Dabiri and Heaslip, 2018; Wu et al., 2018). It outputs 0 when $x_{norm} < 0$, and conversely, it outputs a linear function when $x_{norm} \geq 0$. A visual representation is shown in Fig. 5.

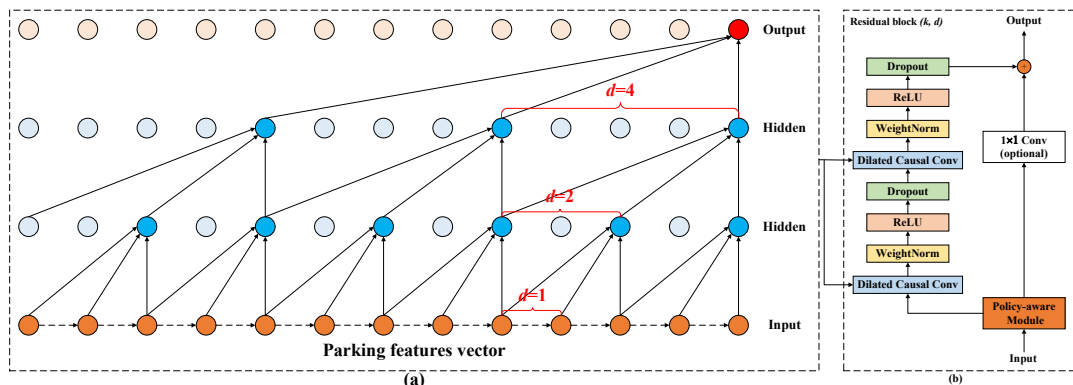


Fig. 4. Structure of the proposed P-TCN model.

Notably, the input X is a 3D vector, and the output of the residual block may have a different number of channels. Thus, the use of a 1×1 convolutional layer on the shortcut branches corrects this issue, ensuring that the input and output have the same width.

3.4. Experimental settings

We utilized a 10-fold cross-validation method to maximize the utilization of the available datasets for training and measure the generalization ability of the model on independent datasets to validate the performance of the proposed P-TCN. The entire dataset was split into two parts with 70% of the data used as the training set for model training, hyperparameter tuning, and validation. The remaining 30% of the data were reserved as the test set to evaluate the prediction accuracy of the final model.

It is worth noting that the 10-fold cross-validation method is a common approach in neural networks for parameter-tuning optimization. As shown in Fig. 6, the training set was further divided into ten equal-sized subsets. The model was trained ten times, with each training using nine subsets as the training set and the remaining subset as the validation set. During each cross-validation, the model was trained using the training data and evaluated using the validation data to determine the best model parameters. After performing 10 cross-validations on the training set and obtaining the best model parameters, these parameters were used in the test set to evaluate the model's generalization ability on independent datasets. Using the 10-fold cross-validation method, we can effectively train better-performing neural network models and avoid overfitting and underfitting.

In addition, three commonly used error metrics, namely, the mean absolute error (MAE), mean absolute percentage error (MAPE), root-mean-square error (RMSE), and fit metric (R^2), were used to evaluate the predictive performance of the proposed P-TCN predictor (Li et al., 2021; Wu et al., 2018; Zheng et al., 2021), which is mathematically defined as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

$$MAPE = 100\% \times \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{\sum_{i=1}^n y_i} \quad (6)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

After conducting a methodological comparison, the superior performance of the P-TCN model was verified. We chose several widely used architectures, including MLR, ARIMA, FCN, LSTM, and TCN, as baselines to compare the data prediction results of our proposed model. In addition, a crucial task was to determine whether the COVID-19 pandemic response caused differences in the collected data. The P-TCN model includes a policy-aware module that enables the identification of policy differences in data.

To further evaluate the performance of our proposed P-TCN model and gain insights into the underlying factors driving parking occupancy, SHAP analysis was conducted on the test set. SHAP values provide a way to explain the output of any machine learning model and can help identify the most important features for predicting parking occupancy (Lundberg and Lee, 2017).

4. Results and discussion

The proposed P-TCN model was used to train and predict 3D data consisting of policy attribute features, time steps, and residential parking occupancy-related features. The P-TCN model employed a policy-aware module to analyze the differences in data under different policy attributes, allowing for more accurate predictions. The model estimation process involved comparing the prediction performance of the proposed P-TCN model with that of other comparison models, including MLR, ARIMA, FCN, LSTM, and TCN models. These models were evaluated on both regular and irregular time-series datasets to determine their effectiveness in predicting residential parking occupancy-related data. Based on the analysis and discussion of the results, suggestions were provided to managers regarding the use of these models for predicting and managing parking occupancy.

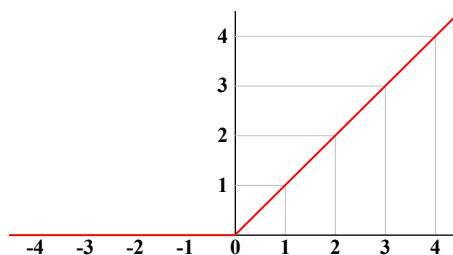


Fig. 5. The ReLU activation function.

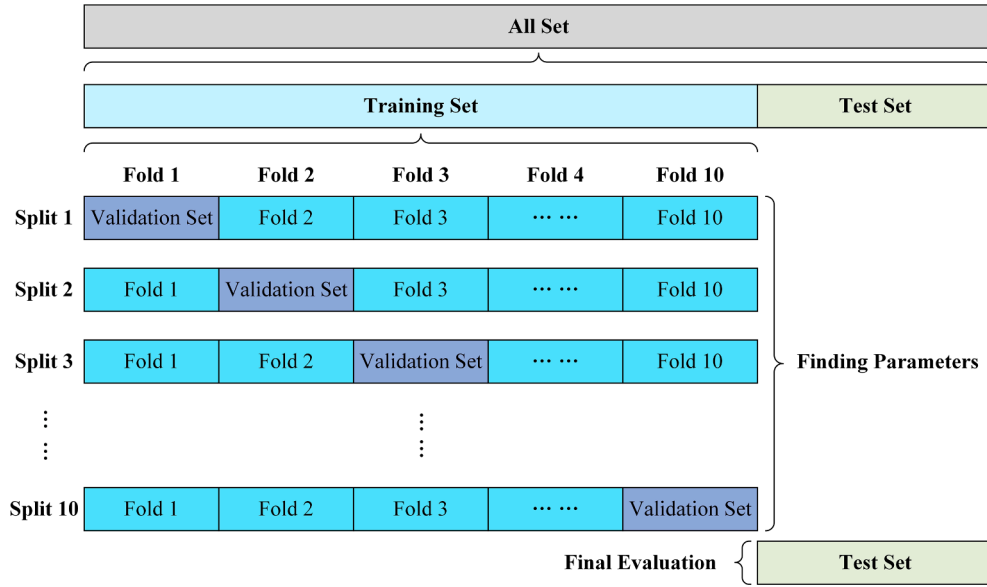


Fig. 6. 10-fold cross-validation approach.

4.1. Policy difference analysis results

To analyze the differences in parking occupancy data, the LSD method was used, and a normal distribution test was performed using the Kolmogorov-Smirnov method to ensure accuracy. The results indicated that the sample data conformed to a normal distribution (P value = 0.55 > 0.05), providing a reliable basis for subsequent analysis. Notably, only significance levels < 0.05 indicate that the differences are significant.

Post hoc multiple comparisons were performed using the LSD method to analyze the differences in parking occupancy data between each weekday and each day of the weekend, as shown in Table 3. The results showed that the variability of the data between each weekday was insignificant ($p > 0.05$), as was the same for each weekend day ($p > 0.05$). However, a significant difference was found between the data for each weekday and weekend day ($p < 0.05$). Therefore, the parking occupancy data were divided into two categories for further analysis: weekdays and weekends.

Parking behavior was classified into weekdays and weekends. This classification is more practical than dividing the day into different periods or the year into different seasons because there are many differences in parking behavior between weekdays and weekends, which affect the amount and distribution of parking demand (Mo et al., 2021). Dividing the day into different time periods

Table 3

Analysis of the differences for each day of the week.

Target groups	Contrast groups	Mean deviation	Standard error	P-value
Monday	Tuesday	-0.0001	-0.002	1.000
	Wednesday	-0.0025	-0.011	0.995
	Thursday	-0.0039	-0.018	0.943
	Friday	-0.0101	-0.043	0.101
	Saturday	0.0497	0.213	0.000
	Sunday	0.0692	0.297	0.000
Tuesday	Wednesday	-0.0024	-0.010	0.995
	Thursday	-0.0039	-0.017	0.947
	Friday	-0.010	-0.043	0.105
	Saturday	0.0497	0.213	0.000
	Sunday	0.0693	0.297	0.000
Wednesday	Thursday	-0.0014	-0.006	1.000
	Friday	-0.0076	-0.032	0.398
	Saturday	0.0521	0.224	0.000
	Sunday	0.0717	0.307	0.000
Thursday	Friday	-0.0061	-0.026	0.656
	Saturday	0.0536	0.230	0.000
	Sunday	0.0731	0.314	0.000
Friday	Saturday	0.0597	0.256	0.000
	Sunday	0.0793	0.340	0.000
Saturday	Sunday	0.0195	-0.014	0.357

presents many challenges, such as defining the start and end times of each time period and balancing the amount of data between time periods (Gragera and Albalade, 2016; Wang et al., 2020). In addition, the relatively slow seasonal changes during the year may neglect short-term changes in parking demand. Therefore, dividing parking behavior into weekdays and weekends can better reflect the actual situation and more accurately analyze and forecast parking occupancy.

Further analysis was conducted to compare the detailed differences in data during different pandemic responses based on the weekday and weekend categories. The results in Table 4 indicate that the differences in parking behavior during weekdays under low- and mid-risk policies are insignificant ($p = 0.804 > 0.05$). However, the differences between low- and mid-risk policies on parking behavior during weekends were highly significant ($p = 0.036 < 0.05$). Parking behavior for high-risk policies differed significantly ($p = 0.000 < 0.05$) from that for low- and mid-risk policies, whether on weekdays or weekends.

Based on the identification of the effect of the pandemic response on data differences, the data were categorized and coded as default variables for input into the P-TCN model.

4.2. Parameter influence on model performance

The choice of hyperparameters is essential for achieving better learning performance and prediction accuracy with neural networks. Therefore, we used well-known and accepted TCN frameworks to select a set of hyperparameters for the proposed P-TCN model (Bai et al., 2018). To minimize the discrepancy (termed 'loss') between the model predictions and actual results in the proposed P-TCN model, we used the Adam optimizer with an initial learning rate of 0.002, which is a widely used value in the literature (Liang et al., 2022b). Another critical hyperparameter is the batch size, which was set to 128 to balance the speed of model training and its generalization ability, following common practices in the literature (Chen et al., 2020). We also set the dropout coefficient to 0.1 in the P-TCN prediction module to prevent overfitting during training and accelerate model convergence (Zhang et al., 2021d).

To achieve accurate predictions, it is crucial to select appropriate values for two important parameters, namely, the number of layers and kernel size, when designing the P-TCN model (Pak et al., 2020; Zhang et al., 2020a). The heatmap presented in Fig. 7 illustrates the combined effects of these two parameters on the predictive performance of the model evaluated using the RMSE. Based on the results, the optimal combination of the number of layers and kernel size for evaluating parking occupancy prediction performance using the RMSE is 4 and 5, respectively.

4.3. Prediction results of parking space occupancy

In practice, evaluating the generalization performance of the proposed P-TCN model is crucial because it can guide the selection of the model and provide a measure of the quality of the final chosen model. In this section, a customary 10-fold cross-validation is used to determine the optimal parameters of the proposed P-TCN model.

The performance of the proposed P-TCN model in predicting parking occupancy was then tested on the test set, and Fig. 8 shows the predicted values and the corresponding observed values. The predicted trends in residential parking occupancy and the observations were consistent, showing the feasibility of the proposed P-TCN model in capturing temporal and anti-pandemic policy changes. Additionally, the model continues to exhibit favorable predictive performance during periods of recurring COVID-19 cases (e.g., December 2021, January 2022, March 2022, and April 2022). However, as shown in Fig. 8 (c), there is a significant error between the actual and predicted values on January 31, 2022, which is Chinese New Year's Day. This can be attributed to people staying at home and celebrating, resulting in a considerable increase in residential parking occupancy and a relatively large error in the forecast results.

Parking occupancy prediction models, including MLR, ARIMA, FCN, LSTM, TCN, and the proposed P-TCN, were compared on the test set. These models were evaluated on regular and irregular datasets, and their MAE, RMSE, MAPE, and R^2 values were compared (Table 5). The results show that the comparison group models, such as FCN, LSTM, and TCN, achieved good fits with R^2 values of 0.832, 0.865, and 0.976, respectively, for predicting regular datasets. However, the comparison group models did not perform satisfactorily when the parking occupancy data were irregular. In contrast, the proposed P-TCN model showed excellent performance on the irregular dataset with reductions of at least 22.86%, 21.43%, and 9.08% in MAE, RMSE, and MAPE prediction errors, respectively. Based on the R^2 values, the prediction accuracy of the P-TCN model was improved by at least 13.80% compared to that of the comparison model. Furthermore, the neural network models (FCN, LSTM, TCN, and P-TCN) outperformed the MLR and ARIMA methods on both regular and irregular datasets.

Table 4
LSD multiple comparisons.

Category	Target policies	Contrast policies	Mean deviation	Standard error	P-value
Weekdays	Low-risk	Mid-risk	-0.007	0.028	0.804
		High-risk	-0.241	0.028	0.000
	Mid-risk	High-risk	-0.234	0.034	0.000
Weekends	Low-risk	Mid-risk	-0.040	0.019	0.036
		High-risk	-0.165	0.019	0.000
	Mid-risk	High-risk	-0.125	0.023	0.000

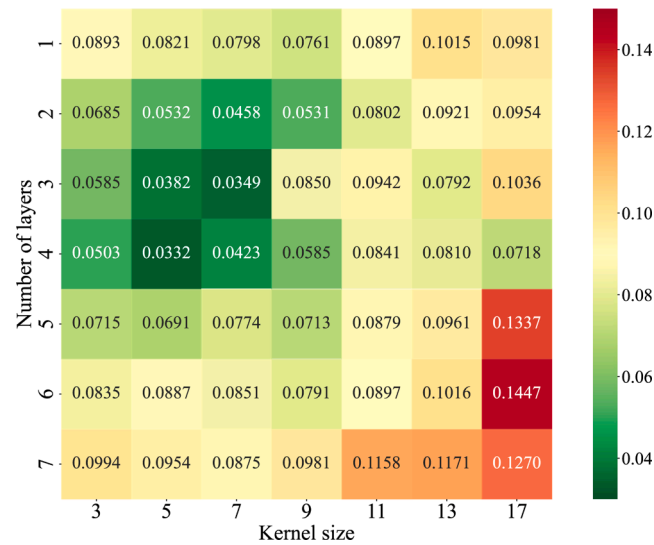


Fig. 7. The combined effect of parameters on model prediction performance.

4.4. Impacts of feature variables on parking policy

Through an in-depth investigation of SHAP values and their application in interpreting and predicting parking occupancy, we reveal the relative importance of different feature variables and their impact on parking policy, such as “policy”, “weather” and “time information”. This approach offers a holistic viewpoint for understanding how these factors influence the prediction of parking occupancy rates. In Fig. 9, the SHAP values of the features are presented, with Fig. 9(a) displaying the global importance of the features arranged in descending order of the average absolute value of the SHAP. Notably, SHAP values are a relative metric; therefore, their interpretation should be compared with the SHAP values of other features (Cai et al., 2022). Fig. 9 (b) depicts a density scatter plot of the features, where larger values of the explanatory variable are indicated in red.

The results reveal that “policy” is the most influential feature in predicting parking occupancy, which is widely accepted by previous studies (Zhang et al., 2022a; Zhang et al., 2021a) and verified by predicted effects and SHAP values (0.75). The importance of identifying the “policy” variable for predicting parking occupancy in residential areas is shown by comparing regular and irregular datasets. The red dots for “policy” are widely distributed in areas with positive SHAP values, indicating that the COVID-19 pandemic response has a significant positive impact on the model prediction. Specifically, as the “policy” variable shifts from a low-risk policy to a high-risk policy, the corresponding parking occupancy rate increases, which is consistent with similar policies implemented in countries such as the United States, France, Australia, and Italy during their COVID-19 anti-pandemic processes (Chan et al., 2021; Zhang et al., 2021a). The outbreak of the COVID-19 pandemic and implementation of the pandemic response have had a dual impact on parking and travel behavior in residential areas. First, it has led to changes in residents’ travel preferences and modes, such as fewer trips and increased preference for self-driving. Second, the pandemic response has affected parking occupancy not only in terms of quantity but also in terms of the allocation of parking time in parking facilities. Therefore, data variability may be a significant barrier to model predictions when influenced by different pandemic responses, as observed in previous studies (Dai et al., 2021; Muren et al., 2022).

Policymakers can use the findings of this study to better understand the impact of the pandemic response on parking and travel behavior and develop a more precise pandemic response based on the level of pandemic risk. This could involve implementing targeted parking and travel restrictions to control the movement of people and minimize the spread of the virus. Moreover, the P-TCN method proposed in this study can be applied to the prediction and interpretation of parking occupancy in different countries and regions, especially in the face of unexpected uncertainty events such as the COVID-19 pandemic, with important application prospects.

The weather-related features, “temperature”, “snowfall”, “humidity”, and “precipitation”, with SHAP values of 0.46, 0.26, 0.23, and 0.11, respectively, have a considerable impact on the prediction. Specifically, “temperature” and “humidity” promote an increase in parking occupancy, while “snowfall” and “precipitation” cause a decrease in parking occupancy. Previous studies on parking occupancy prediction (Fokker et al., 2022; Yang et al., 2019) have also found that in low-temperature and snowy/rainy weather, travelers are more likely to choose to self-drive to avoid the inconvenience and long wait times associated with public transportation, which can lead to increased traffic congestion. Conversely, in winter conditions with high humidity, public transportation may be preferred to avoid inconvenience and safety hazards caused by low visibility due to fog and haze. Similar findings were obtained by Zhou et al. (2017).

These findings provide valuable references for traffic management and for travelers to address traffic congestion under different weather conditions. Traffic managers should improve the convenience of public transportation by increasing the frequency of bus services and providing more real-time traffic information to reduce travelers’ waiting times. Transportation safety reminders and

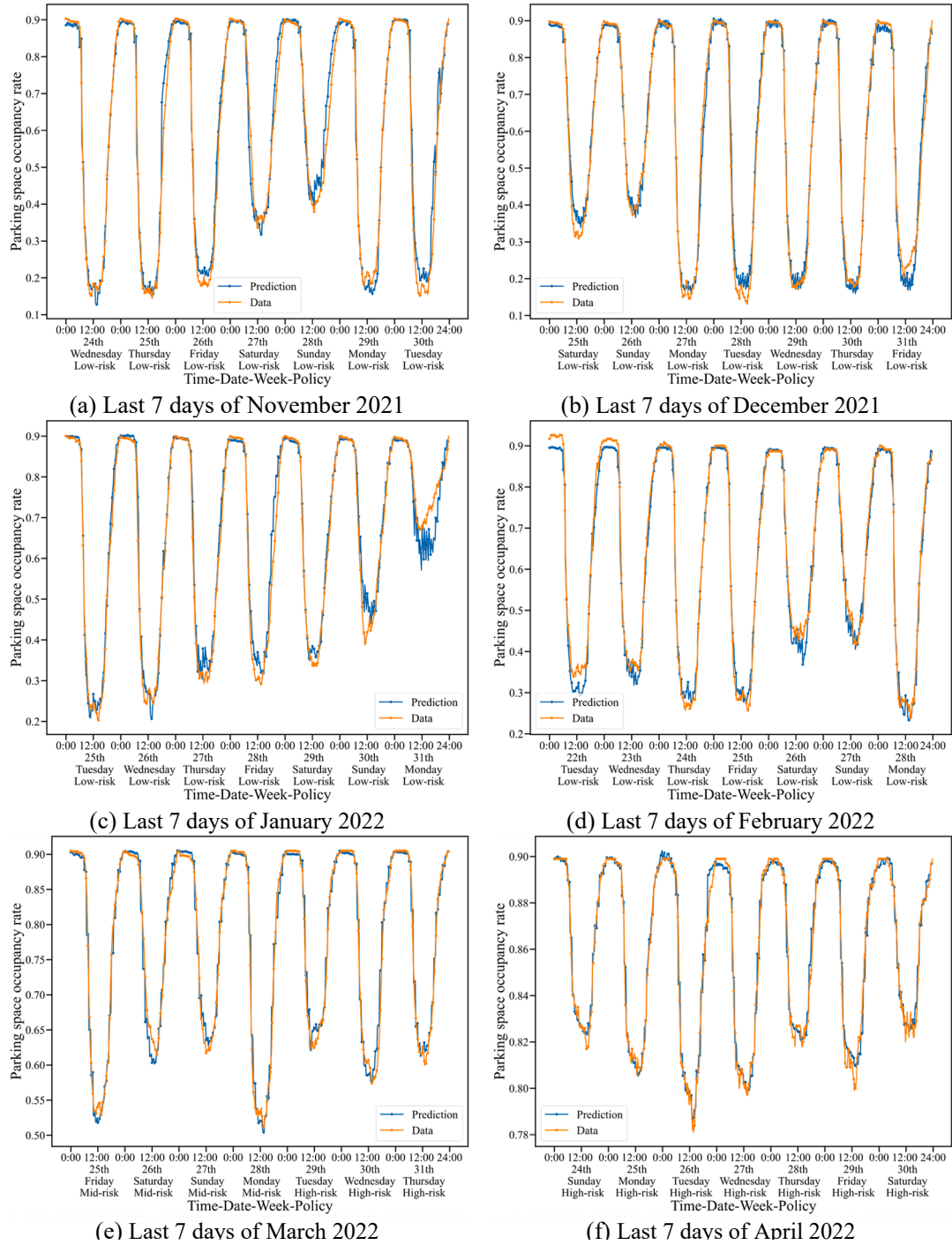


Fig. 8. Prediction results on the test set.

traffic restrictions should be promptly released under adverse weather conditions.

Additionally, parking occupancy is influenced by high-level temporal features such as time of day, day of week, month, season, and holiday, showing cyclical features. Among them, “week”, “time” and “month” have SHAP values of 0.41, 0.31, and 0.15, respectively, which have a significant positive impact on predicting parking occupancy. In particular, the “week” feature reflects the effect of the day of the week on parking occupancy, consistent with the findings of Yang et al. (2022). The “time” feature reflects the impact of periods on parking occupancy, e.g., parking occupancy is relatively lower in the morning and evening peak periods and may be higher in the evening and early morning, consistent with earlier studies (Yang et al., 2019). The feature “month” also showed a significant

Table 5
Model performance on the test set.

Data	Model	Training Dataset				Testing Dataset			
		MAE	RMSE	MAPE	R ²	MAE	RMSE	MAPE	R ²
Regular datasets	MLR	0.062	0.076	16.526%	0.754	0.082	0.087	20.225%	0.562
	ARIMA	0.053	0.063	12.651%	0.793	0.071	0.066	18.225%	0.607
	FCN	0.029	0.015	2.903%	0.997	0.048	0.064	10.445%	0.832
	LSTM	0.023	0.028	3.433%	0.988	0.039	0.052	8.023%	0.865
	TCN	0.012	0.024	5.002%	0.989	0.033	0.036	6.620%	0.976
Irregular datasets	P-TCN	0.011	0.027	6.327%	0.983	0.025	0.029	6.312%	0.981
	MLR	0.113	0.132	24.329%	0.583	0.131	0.095	27.628%	0.431
	ARIMA	0.088	0.107	19.618%	0.719	0.126	0.082	21.554%	0.474
	FCN	0.040	0.056	6.316%	0.828	0.055	0.069	17.937%	0.641
	LSTM	0.028	0.041	5.987%	0.984	0.047	0.058	11.918%	0.745
	TCN	0.019	0.028	5.879%	0.980	0.035	0.042	7.060%	0.855
	P-TCN	0.016	0.023	5.453%	0.981	0.027	0.033	6.419%	0.971

Note: To ensure the reliability of the MLR model, chi-square tests and Pearson correlation tests were conducted, revealing that the feature variables were significantly correlated with the target variables (p -value < 0.05) and needed to be considered in the modeling process. The F-statistic test revealed that MLR does not describe the nonlinear relationship between the data well (p -value > 0.05), and Box-Cox and dummy variable transformations of the variables were necessary to adapt the data to the linear regression model for prediction. Finally, the Wald test was used to test the random parameters to determine whether the feature variable has a significant impact on the target variable (p -value < 0.05).

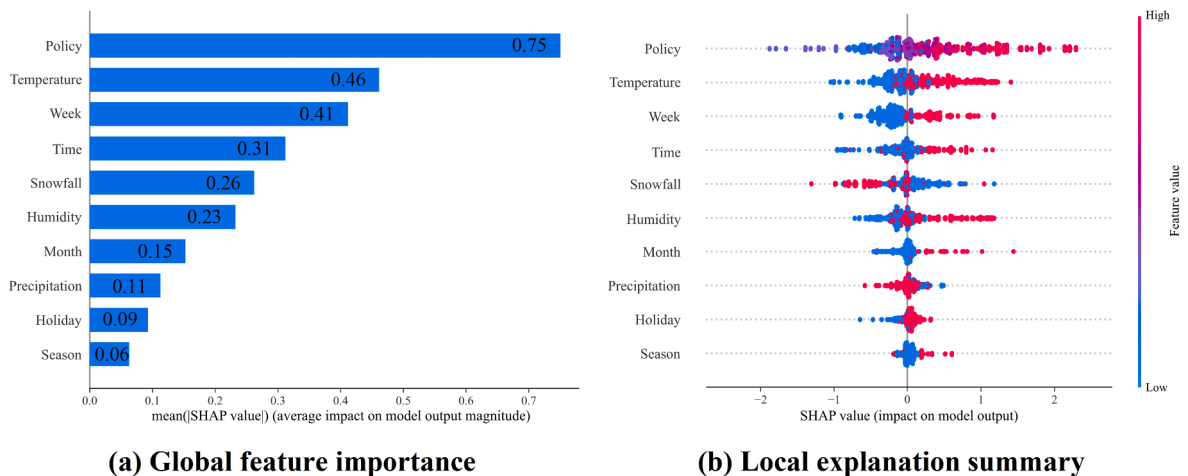


Fig. 9. Results of the SHAP values of features.

positive effect on parking occupancy. In contrast, the contribution of “holiday” and “season” features to the forecast may be small, with SHAP values of 0.09 and 0.06, respectively. Nonetheless, their influence still needs to be considered to reflect trends in parking demand across holidays and seasons, similar to the study by [Arjona Martinez et al. \(2021\)](#).

Research on the time information features of residents’ parking and travel will help better understand the rules of residents’ parking and travel and formulate more effective traffic management policies and measures to improve the efficiency of urban traffic operations and reduce traffic congestion. Traffic managers should encourage residents to arrange their travel times reasonably, strengthen road traffic management, and promote shared travel modes. Public transportation should be encouraged to reduce the flow of private vehicles by offering price discounts during peak periods.

5. Conclusions

This study aims to develop a P-TCN modeling framework for predicting residential parking occupancy using over 52,000 residential parking occupancy records. The prediction results of the MLR, ARIMA, FCN, LSTM, and TCN models were compared, and the contribution of the feature variables to the model predictions was evaluated in conjunction with the SHAP analysis. The results show that the P-TCN model can significantly improve the accuracy of residential parking occupancy forecasting, even in the face of data irregularities caused by different pandemic responses. This framework can help bridge the gap in exploring residential parking occupancy prediction during emergencies such as the COVID-19 outbreak in China and has important application prospects in different countries and regions. The P-TCN model can also serve as decision support for governments and urban planners in predicting residential parking demand and travel behavior, providing references for more accurate transportation planning and policies to improve transportation efficiency and convenience in cities.

Several important findings of this study are closely related to the prediction results of the model. First, “policy” is a critical feature that influences the prediction of residential parking occupancy. The differences in the COVID-19 pandemic response have an impact on parking occupancy and the distribution of parking time in residential areas, as reflected in irregular variations in parking occupancy rate data. Accordingly, research on the impact of corresponding pandemic responses on parking and travel behavior can help develop more precise policies and measures to control population movements and reduce the possibility of virus transmission. Second, weather conditions have a significant impact on parking occupancy. Rain, snow, or cold weather can affect the normal operation of public transportation, resulting in more people choosing to travel by car and increasing traffic congestion. During winter with high humidity, public transportation or taxi rides may be preferred to avoid inconvenience and safety hazards caused by weather conditions, such as fog and haze. Increasing the frequency of public transportation and providing real-time traffic information can help to reduce traffic congestion and improve traffic safety. Furthermore, under severe weather conditions, it is necessary to issue safety warnings and impose traffic restrictions. Finally, the time of day, day of the week, month, and other temporal features have an impact on the change in parking occupancy, showing a cyclical trend. Understanding the impact of these temporal features on parking and travel patterns can help to adopt more effective traffic management policies and measures to improve the efficiency of urban traffic operations and reduce traffic congestion. Transportation managers play a role in encouraging residents to use public transportation during peak hours and in reducing private vehicle traffic by promoting shared mobility modes and improving road traffic management.

This study has several potential limitations that need to be addressed. First, due to the limitations of data collection and insufficient learning of “holiday” and “season” features, the contribution of these two variables to the prediction model needs improvement. Future research should collect and analyze more data on “holiday” and “season” features to improve the model’s understanding of these factors. Second, only the effects of factors such as policy, weather, and temporal information on parking occupancy were considered without considering the possible effects of other variables such as land use, density, car ownership, public transport, and parking prices on parking demand. More variables should be considered in future studies to analyze changes in parking demand more comprehensively. Finally, while SHAP analysis was employed in this study to assess the contribution of feature variables to model predictions, the analysis may be subject to the limitations and interpretative challenges of the model itself. Therefore, further exploration and analysis of the interpretation of the model prediction results are needed. In addition, future research should explore the application of shared parking strategies within the P-TCN modeling framework. The modeling and prediction of shared parking usage frequency, parking duration, and user behavior and the analysis of the synergistic relationship between shared parking and other travel modes can promote the sustainable development of urban transportation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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